

| Question | Answer          | Mark | Mark scheme                                                                                                                                               |
|----------|-----------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22       | 12 red, 9 green | P1   | for process to find a relationship between $r$ and $g$<br>eg $\frac{g}{r+g} = \frac{3}{7}$ <b>or</b> $\frac{g}{r} = \frac{3}{4}$                          |
|          |                 | P1   | for process to find a second relationship between $r$ and $g$<br>eg $\frac{g+3}{r+2+g+3} = \frac{6}{13}$ <b>or</b> $\frac{g+3}{r+2} = \frac{6}{7}$        |
|          |                 | P1   | (dep P2) for start to process of solving pair of equations, eg eliminates one variable from the equations <b>or</b> removes fractions from both equations |
|          |                 | P1   | (dep P3) for complete process to solve equations to find $g$ <b>or</b> $r$                                                                                |
|          |                 | A1   | cao<br><br><b>OR</b>                                                                                                                                      |
|          |                 | P1   | for two of $3x + 3$ , $4x + 2$ and $7x + 5$                                                                                                               |
|          |                 | P1   | for $\frac{3x+3}{7x+5} = \frac{6}{13}$                                                                                                                    |
|          |                 | P1   | (dep P2) for removing fractions from the equation,<br>eg $13(3x + 3) = 6(7x + 5)$ <b>or</b> $39x + 39 = 42x + 30$                                         |
|          |                 | P1   | (dep P3) for complete process to solve $13(3x + 3) = 6(7x + 5)$                                                                                           |
|          |                 | A1   | cao                                                                                                                                                       |